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A Novel Self-Sustaining Process in Plane Couette Flow

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Abstract

This thesis investigates the existence of a novel, spanwise-driven self-sustaining process (SSP) in turbulent plane Couette flow through the application of a spanwise quasilinear approximation (QL_z). By restricting the low-wavenumber subspace exclusively to the spanwise mean, the classical streamwise turbulence maintenance pathway is mathematically suppressed, and a new mechanism for turbulence sustenance emerges. Simulations are conducted at a bulk Reynolds number of $Re_w = 1428$, which successfully sustains a turbulent state with a friction Reynolds number of $Re_\tau = 63$. Spectral energy analyses and targeted mode-truncation experiments reveal an alternative regenerative cycle governed by a distinct set of triadic interactions. In this novel SSP, the oblique modes $(\alpha, \pm\beta)$ and spanwise streaks $(0, \beta)$ extract kinetic energy directly from the $(0, 0)$ mean shear. The nonlinear scattering of these modes generates the Reynolds stress that energises the fundamental wavy spanwise-mean structure $(\alpha, 0)$. The subsequent linear instability and structural breakdown of this wavy base flow directly excite the high-wavenumber fluctuations and vortices which allow the oblique modes and spanwise streaks to extract energy from the mean flow and thereby closing the loop. These findings demonstrate that wall-bounded turbulence can self-sustain via a fundamentally distinct pathway, completely bypassing the canonical reliance on streamwise vortex regeneration.

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Nomenclature

Acronyms

Acronym	Definition
CFL	Courant–Friedrichs–Lewy
DNS	Direct Numerical Simulation
GQL	Generalised Quasilinear
QL	Quasilinear
QL_x	Quasilinear approximation based on the streamwise mean
QL_z	Quasilinear approximation based on the spanwise mean
RMS	Root-Mean-Square
SSP	Self-Sustaining Process

Roman Symbols

Symbol	Definition
E_i	Kinetic energy of the oblique mode (α, β)
E_s	Kinetic energy of the spanwise streaks $(0, \beta)$
E_x	Kinetic energy of the fundamental wavy spanwise-mean mode $(\alpha, 0)$
$\hat{E}(\mathbf{k})$	Spectral kinetic energy of mode $\mathbf{k}$
h	Channel half-height
$\mathbf{k}$	wavevector (k_x, k_z)
L_x, L_y, L_z	Domain dimensions in the streamwise, wall-normal, and spanwise directions

Continued on next page

Roman Symbols (continued)

Symbol	Definition
M_x, M_z	Spectral cutoff wavenumbers defining the low-wavenumber subspace
N_x, N_z	Maximum resolved wavenumbers in the numerical domain
p	Pressure
p_l, p_h	Pressure components for the low- and high-wavenumber subspaces
$\mathcal{P}$	Turbulent kinetic energy production
$\hat{P}$	Spectral turbulent production
$\mathcal{P}_l, \mathcal{P}_h$	Projection operators for the low- and high-wavenumber subspaces
Re_τ	Friction Reynolds number, $u_\tau h / \nu$
Re_w	Bulk Reynolds number, $U_w h / \nu$
$\hat{T}_p$	Spectral pressure transport
$\hat{T}_{\text{turb}}$	Spectral nonlinear turbulent transfer
$\hat{T}_\nu$	Spectral viscous transport
$\mathbf{u}$	Velocity vector (u, v, w)
$\mathbf{U}$	Statistically averaged mean velocity vector
$\mathbf{U}_l$	Low-wavenumber velocity component
$\mathbf{u}_h$	High-wavenumber velocity fluctuation
$u_{i,\text{rms}}$	Root-mean-square velocity fluctuations
u_τ	Friction velocity, $\sqrt{\tau_w / \rho}$
U_w	Wall velocity
V	Volume of the computational domain
x, y, z	Cartesian coordinates in the streamwise, wall-normal, and spanwise directions
$\mathbf{x}_\parallel$	Horizontal position vector (x, z)

Greek Symbols

Symbol	Definition
α	Fundamental streamwise wavenumber, $2\pi/L_x$
β	Fundamental spanwise wavenumber, $2\pi/L_z$
δ_ν	Characteristic viscous length scale, ν/u_τ
$\hat{\epsilon}$	Spectral viscous dissipation
ϵ_{L_2}	Relative L_2 norm error
λ^+	Average spanwise spacing of near-wall streaks
ν	Kinematic viscosity
ρ	Fluid density
τ_w	Mean wall shear stress

Subscripts, Superscripts, and Operators

Symbol	Definition
$(\cdot)^+$	Variable non-dimensionalised by inner scales (wall units)
$\langle \cdot \rangle$	Spatiotemporal or ensemble average
$(\cdot)'$	Fluctuating component in the Reynolds decomposition
$\widehat{(\cdot)}$	Complex Fourier coefficient
$(\cdot)^*$	Complex conjugate

Chapter 1

Introduction

1.1 Motivation

Wall-bounded turbulence is one of the most interesting topics in fluid dynamics, having intrigued researchers since the seminal experiments of Osborne Reynolds in the 19th century. Beyond its intrinsic scientific interest, it has been extensively investigated due to its numerous and crucial applications, ranging from atmospheric dynamics [2] to drag reduction in aerospace vehicles [3]. Fundamental features of wall-bounded turbulence have been documented since the early 20th century, most notably the logarithmic dependence of the mean velocity profile derived by Von Kármán in 1930 [4]. However, the precise mechanisms by which turbulence self-sustains in wall-bounded shear flows remain an active area of research.

1.2 Dynamics of wall-bounded turbulence

The initial theoretical framework for the mechanistic structure of shear turbulence emerged in the mid-20th century. Malkus [5] pioneered the application of stability theory to the mean turbulent field, postulating that the mean velocity profile organises into a marginally stable state that maximises dissipation. Simultaneously, Townsend [6; 7] introduced the attached eddy hypothesis, advancing the structural view of wall turbulence by proposing that the logarithmic region is populated by a hierarchy of self-similar, wall-attached turbulent eddies whose length scales are proportional to their distance from the boundary.

Experimental observations since 1967 have revealed that wall-bounded turbulence is not entirely stochastic, but is instead populated by highly organised coherent structures [8; 9]. The most prominent of these are near-wall streaks, regions of high- and low-speed streamwise momentum alternating in the spanwise direction. The critical breakthrough in understanding how these structures maintain the turbulent state was achieved by Hamilton, Kim, & Waleffe [1], who identified a ‘self-sustaining process’ (SSP) in near-wall turbulence. The SSP relies on a continuous, nonlinear feedback loop involving the linear transient growth of streaks, their subsequent breakdown, and the nonlinear regeneration of quasi-streamwise vortices. Subsequent research has demonstrated that coherent structures at larger scales, up to the order of the outer length scale h , known as large- and very large-scale motions [10], are statistically and dynamically self-similar [11; 12] and independently self-sustain via an SSP loop analogous to that of the near-wall streaks.

1.3 The Quasilinear Framework

Because the primary coherent structures driving turbulence maintenance exist as streamwise or quasi-streamwise independent structures, recent efforts to understand and model the SSP have heavily utilised the Quasilinear (QL) approximation. Originally rooted in the theoretical frameworks of Malkus [5] and Herring [13], and later adapted as Stochastic Structural Stability Theory (SSST) [14] and the Restricted Non-Linear (RNL) model [15], the QL framework divides the flow into the low-wavenumber modes and the rest of the fluctuations. By artificially restricting fluctuation-fluctuation nonlinear scattering, the QL framework allows the fluctuations to evolve linearly about the mean, feeding back into the mean field exclusively via a Reynolds stress.

Standard QL studies, which define the low-wavenumber field strictly as the streamwise mean (QL_x), have shown very close correspondence to the bare Navier-Stokes turbulent state, proving that streak instability excites fluctuating modes, and that their triadic interactions are solely responsible for vortex regeneration and subsequent streak growth.

1.4 Objectives of the Present Study

The classical SSP is inherently based on the dominance of streamwise coherent structures and is mathematically validated by the success of the QL_x approximation. However, this raises a fundamental physical question: is the classical streamwise pathway the exclusive mechanism for turbulence sustenance, or do alternative SSP cycles exist?

This thesis investigates turbulence maintenance in plane Couette flow by applying a novel formulation of the Quasilinear Approximation based exclusively on the spanwise mean flow (QL_z). This study aims to discover whether turbulence can self-sustain through an entirely distinct, spanwise-driven maintenance mechanism. Uncovering such a mechanism would reveal a missing physical component of turbulence maintenance which could be operating alongside the classical SSP.

1.5 Thesis Outline

The thesis is structured as follows. Chapter 2 reviews the theoretical background necessary to understand its results, including spectral energy budgets, the classical self-sustaining process (SSP), and the mathematical foundations of the quasilinear (QL) approximation. Chapter 3 outlines the numerical methodology, describing the solver, grid resolutions, and the implementation of the quasilinear framework. Chapter 4 presents the numerical configuration and validates the solver against established literature databases. Chapter 5 analyses the statistical characteristics of turbulence maintained under the QL_z framework. Chapter 6 investigates the underlying dynamics of the spanwise-driven SSP. Chapter 7 discusses the physical implications of the spanwise-driven self-sustaining process, presenting a conceptual schematic of it, and finally Chapter 8 summarises the principal findings of the study and outlines potential directions for future research.

Chapter 2

Background

2.1 Wall-Bounded Turbulence and Plane Couette Flow

2.1.1 Governing Equations and Flow Configuration

The flow configuration considered in this study is turbulent plane Couette flow, the canonical shear-driven flow bounded by two infinite parallel plates, which in practice are truncated in the homogeneous directions to a periodic computational box of dimensions $L_x \times 2h \times L_z$. A Cartesian coordinate system (x, y, z) is adopted, aligning with the streamwise, wall-normal, and spanwise directions, respectively. The plates, located at $y = \pm h$, move in opposite directions with velocities $\pm U_w$, and the system is subject to no-slip and no-penetration boundary conditions at the walls, yielding $\mathbf{u}(x, \pm h, z, t) = (\pm U_w, 0, 0)$. Figure 2.1 shows a diagram of the domain used throughout the present study.

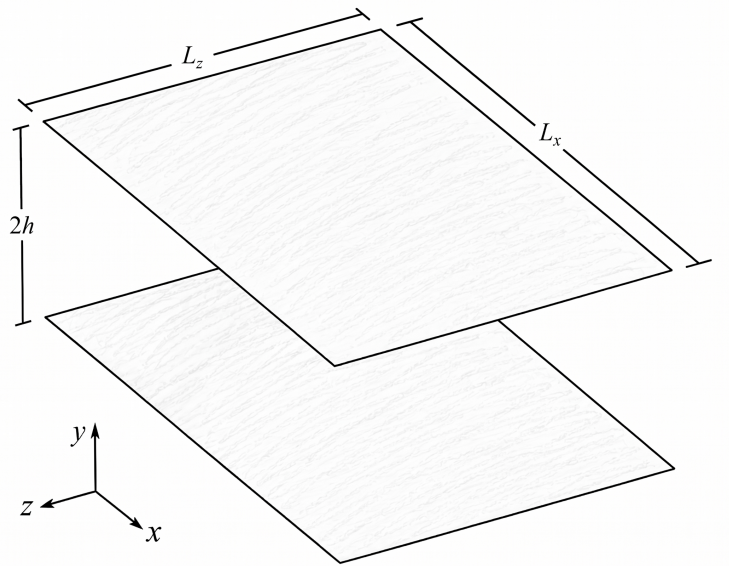


Figure 2.1: Diagram of flow geometry with coordinate conventions used in the present study.

The fluid is assumed to be incompressible, with constant density ρ and kinematic viscosity ν . The flow is strictly governed by the incompressible Navier-Stokes and continuity equations, where $\mathbf{u} = (u, v, w)$ is the velocity vector and p is the pressure.

The system is characterised by two distinct Reynolds numbers. The macroscopic state of the flow is defined by the bulk Reynolds number, $Re_w = U_w h / \nu$, and the turbulent state near the

boundary is characterised by the friction Reynolds number, $Re_\tau = u_\tau h / \nu$. Here, $u_\tau = \sqrt{\tau_w / \rho}$ is the friction velocity, driven by the mean wall shear stress $\tau_w = \rho \nu \left. \frac{d\langle u \rangle}{dy} \right|_{wall}$, where $\langle \cdot \rangle$ denotes an appropriate ensemble or spatial average. The characteristic viscous length scale is defined in wall units as $\delta_\nu = \nu / u_\tau$. Throughout this thesis, variables non-dimensionalised by the inner scales u_τ and δ_ν are denoted by the superscript $(\cdot)^+$.

To characterise the turbulent state, the instantaneous velocity field is separated into a statistically averaged mean and a fluctuating component via the Reynolds decomposition, $\mathbf{u}(\mathbf{x}, t) = \mathbf{U}(y) + \mathbf{u}'(\mathbf{x}, t)$. For statistically steady plane Couette flow, averaging over time and the homogeneous wall-parallel directions (x, z) yields a mean velocity profile strictly in the streamwise direction, $\mathbf{U} = (U(y), 0, 0)$.

The intensity of the turbulent fluctuations is quantified by the root-mean-square (RMS) of the velocity components, defined as:

$$u_{i,rms} = \sqrt{\langle u_i'^2 \rangle}, \quad (2.1)$$

where $\langle \cdot \rangle$ denotes the spatiotemporal average. These velocity variances provide a direct, measurable footprint of the coherent structures governing the flow, and are used as a standard benchmark for verifying that a numerical domain properly captures the mechanisms of the SSP.

By exploiting the symmetries of the domain and integrating the streamwise momentum equation, the mean momentum balance for plane Couette flow is obtained. It can then be normalised by the inner scales to get:

$$\frac{dU^+}{dy^+} - \langle u'^+ v'^+ \rangle = 1. \quad (2.2)$$

Here, the transport of momentum across the channel is governed by the Reynolds shear stress, $-\langle u'v' \rangle$. This term is of paramount physical importance because it dictates the rate of turbulent kinetic energy production, $\mathcal{P} = -\langle u'v' \rangle \frac{dU}{dy}$, which represents the dominant term responsible for the transfer of energy from the mean flow to the turbulent fluctuations.

2.1.2 Spectral Energy Budget and Triadic Interactions

As the flow is homogeneous in the streamwise and spanwise directions, the velocity field can be expanded into a spatial Fourier series:

$$\mathbf{u}(\mathbf{x}_\parallel, y, t) = \sum_{\mathbf{k}} \hat{\mathbf{u}}(\mathbf{k}, y, t) e^{i\mathbf{k} \cdot \mathbf{x}_\parallel} \quad (2.3)$$

where we defined the horizontal position vector $\mathbf{x}_\parallel = (x, z)$ and the corresponding 2D wavevector $\mathbf{k} = (k_x, k_z)$, and $\hat{\mathbf{u}}$ denotes the complex Fourier coefficients. To understand how different spatial scales interact, we substitute this expansion into the nonlinear convective term of the Navier-Stokes equations. Assigning the dummy wavevector $\mathbf{p}$ to the velocity mode and $\mathbf{q}$ to the velocity

gradient mode, the product in physical space becomes the multiplication of two series:

$$\begin{aligned} (\mathbf{u} \cdot \nabla) \mathbf{u} &= \sum_{\mathbf{p}} \hat{\mathbf{u}}(\mathbf{p}, y, t) e^{i\mathbf{p} \cdot \mathbf{x}_{\parallel}} \cdot \sum_{\mathbf{q}} (\widehat{\nabla \mathbf{u}'})(\mathbf{q}, y, t) e^{i\mathbf{q} \cdot \mathbf{x}_{\parallel}} \\ &= \sum_{\mathbf{p}} \sum_{\mathbf{q}} \left[\hat{\mathbf{u}}(\mathbf{p}, y, t) \cdot (\widehat{\nabla \mathbf{u}'})(\mathbf{q}, y, t) \right] e^{i(\mathbf{p}+\mathbf{q}) \cdot \mathbf{x}_{\parallel}}. \end{aligned} \quad (2.4)$$

Transforming the governing Navier-Stokes equations into Fourier space and isolating the governing equation for a specific target mode $\mathbf{k}$ reveals the fundamentally mechanistic nature of the nonlinear convective term. Due to the orthogonality of complex exponentials, the double sum above will only contribute to the target mode $\mathbf{k}$ when the argument of the exponential matches $\mathbf{k} \cdot \mathbf{x}_{\parallel}$. This mathematical projection demonstrates that energy is transferred between spatial scales exclusively through triadic interactions [16]. A velocity mode with wavevector $\mathbf{p}$ interacts with a velocity gradient mode $\mathbf{q}$ to force a third mode $\mathbf{k}$, subject to the strict geometric triad condition:

$$\mathbf{p} + \mathbf{q} = \mathbf{k}. \quad (2.5)$$

We will see in Section 2.2 that specific triadic interactions are the engine of the SSP.

To quantify the dynamical maintenance of a specific flow structure, one must examine the spectral kinetic energy budget. By taking the inner product of the Fourier-transformed momentum equation with the complex conjugate velocity $\hat{\mathbf{u}}^*(\mathbf{k})$ and integrating across y , the rate of kinetic energy transfer into a specific mode, defined as $\hat{E}(\mathbf{k}) = \frac{1}{2} \hat{\mathbf{u}}^*(\mathbf{k}) \cdot \hat{\mathbf{u}}(\mathbf{k})$, is derived as follows [17; 18]:

$$\frac{\partial \hat{E}(\mathbf{k}, y)}{\partial t} = \hat{P}(\mathbf{k}, y) - \hat{\varepsilon}(\mathbf{k}, y) + \hat{T}_{turb}(\mathbf{k}, y) + \hat{T}_p(\mathbf{k}, y) + \hat{T}_\nu(\mathbf{k}, y) \quad (2.6)$$

where the terms on the right-hand side represent turbulent production ($\hat{P}$), viscous dissipation ($\hat{\varepsilon}$), nonlinear turbulent transfer ($\hat{T}_{turb}$), pressure transport ($\hat{T}_p$), and viscous transport ($\hat{T}_\nu$). Previous studies [19] have shown that while redistribute energy in the wall-normal direction, the overall contribution of pressure and viscous transport to the energy balance of large-scale structures is negligible in the logarithmic and outer regions. Therefore, to understand the energisation of specific modes in Couette flow, our focus narrows to the two primary source mechanisms for a given mode $\mathbf{k}$:

$$\hat{P}(\mathbf{k}, y) = \text{Re} \left\{ -\hat{\mathbf{u}}^*(\mathbf{k}, y) \hat{v}(\mathbf{k}, y) \frac{dU}{dy} \right\} \quad (2.7)$$

$$\hat{T}_{turb}(\mathbf{k}, y) = \text{Re} \left\{ -\hat{\mathbf{u}}^*(\mathbf{k}, y) \cdot \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\hat{\mathbf{u}}(\mathbf{p}, y) \cdot \nabla) \hat{\mathbf{u}}(\mathbf{q}, y) \right\} \quad (2.8)$$

Effectively, $\hat{P}(\mathbf{k})$ represents the spectral density of the turbulent production and hence quantifies the scale-by-scale extraction of kinetic energy from the mean shear by the velocity fluctuations at a specific wavenumber vector $\mathbf{k}$, while $\hat{T}_{turb}$ quantifies how energy is routed from other spatial scales into the target mode $\mathbf{k}$ via the nonlinear triadic interactions defined earlier.

2.2 The Classical Self-Sustaining Process (SSP)

The fundamental mechanisms governing the maintenance of wall-bounded turbulence were codified by Hamilton, Kim, & Waleffe [1], who established the theoretical framework for the canonical SSP. Building upon the identification of near-wall streaks characterised by an average spanwise spacing of $\lambda^+ \approx 100$ [8], Hamilton et al. [1] used direct numerical simulation of plane Couette flow in a minimal flow unit, the smallest possible computational domain capable of autonomously sustaining turbulence. This geometric restriction was deliberately employed to isolate the dynamically significant structures at their fundamental length scales. By constraining the domain size, the fundamental spanwise wavenumber, defined as $\beta = 2\pi/L_z$, maps exactly to the characteristic spacing of a single near-wall streak pair, and the fundamental streamwise wavenumber, $\alpha = 2\pi/L_x$, is structurally forced to act as the principal 3D streak instability mode (α, β) . Their work demonstrated that within this isolated spectral space, turbulence is sustained through a cyclic, three-stage process: the formation of streaks via the linear lift-up effect, the breakdown of these streaks through the 3D instabilities, and the subsequent nonlinear regeneration of streamwise vortices. The process is shown schematically in Figure 2.2 and is described more thoroughly later. Subsequently, turbulent coherent structures which also exist at larger scales, up to lengths of the order of the outer length scale h of the flow, known as large- and very large-scale motions [10], have also been found to be independently sustained via a similar ‘self-sustaining process’ to the near-wall streaks and to be statistically and dynamically self-similar [11; 12].

2.2.1 Lift-up

The lift-up mechanism relies on the transport of momentum by quasi-streamwise vortices. These vortices induce wall-normal velocity fluctuations, v , which advect the background mean shear, $\partial U/\partial y$. This lifts low-speed fluid away from the wall to form low-speed streaks, and sweeps high-speed fluid towards the wall to form high-speed streaks [20; 21]. Mathematically, this is represented in the streamwise momentum equation by the linear production term:

$$-v \frac{\partial U}{\partial y} \quad (2.9)$$

Because the streaks are uniform in the streamwise direction, this process is dominated by modes with streamwise wavenumber $k_x = 0$.

2.2.2 Streak breakdown

Analysing perturbations around a base flow comprising the mean streamwise velocity and the spanwise streaks, Hamilton et al. [1] originally modelled streak breakdown as a modal linear instability. In this framework, the strong inflectional shear of the streaks supports exponentially growing eigenmodes, characterised by wavevectors (α, β) and $(\alpha, 0)$, leading to the physical breakdown and waviness of the streak structures [22; 23]. However, the precise physical mechanism of this breakdown remains heavily debated. Because the linearised Navier-Stokes oper-

ator is highly non-normal, perturbations can undergo rapid transient growth even in the absence of an unstable eigenmode, a mechanism formally known as transient growth [24]. Further studies have argued that typical streak amplitudes in near-wall turbulence are frequently insufficient to trigger classical modal instability, suggesting that streak breakdown is instead driven by the transient, non-modal amplification of optimal perturbations [25].

2.2.3 Vortex-regeneration

The nonlinear scattering of the instability modes, the triad $(\pm\alpha, 0)$ and $(\mp\alpha, \beta)$, produces a direct Reynolds stress forcing at the wavenumber $(0, \beta)$ which drives the regeneration of streamwise vortices, thereby closing the SSP cycle.

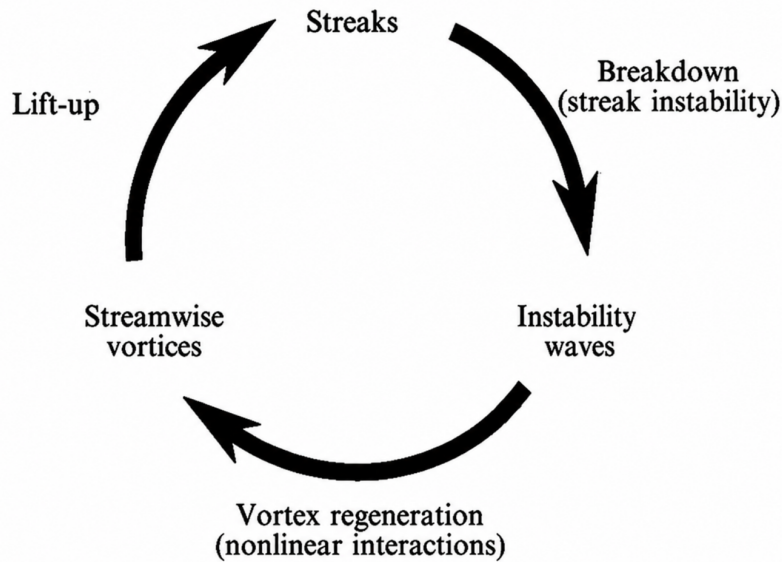


Figure 2.2: Schematic of the classical SSP (Adapted from [1]).

2.3 The Quasilinear (QL) Approximation

The QL approximation, and its expansion into the generalised quasilinear (GQL) framework [26; 18], provides a mathematical framework to isolate and study the triadic interactions responsible for turbulence maintenance.

Mathematical Formulation

In this study, the QL approximation is formally introduced as a limit of the GQL framework [26]. Following the Fourier representation introduced in Section 2.1.2, the velocity field is decomposed into a low-wavenumber component, $\mathbf{U}_l$, and a high-wavenumber fluctuation, $\mathbf{u}_h$, such that $\mathbf{u} = \mathbf{U}_l + \mathbf{u}_h$. The low-wavenumber velocity field is formally defined as:

$$\mathbf{U}_l = \sum_{n=-M_z}^{M_z} \sum_{m=-M_x}^{M_x} \hat{\mathbf{u}}_{m,n} e^{i(m\alpha x + n\beta z)} \quad (2.10)$$

and $\mathbf{u}_h$ is the rest of the fluctuation. Here, $\alpha = 2\pi/L_x$ and $\beta = 2\pi/L_z$ are the fundamental streamwise and spanwise wavenumbers, respectively. The parameters M_x and M_z define the threshold spectral cutoffs. Standard Direct Numerical Simulation (DNS) is recovered in the limit where $M_x = N_x$ and $M_z = N_z$, with N_x and N_z being the maximum resolved wavenumbers in the numerical domain. Conversely, the QL approximations are achieved by setting M_x or $M_z = 0$.

This decomposition is formalised by introducing linear projection operators for the low- and high-wavenumber subspaces:

$$\mathcal{P}_l[\mathbf{u}] \equiv \mathbf{U}_l, \quad \mathcal{P}_h[\mathbf{u}] \equiv \mathbf{u} - \mathbf{U}_l = \mathbf{u}_h \quad (2.11)$$

which satisfy the following properties:

$$\mathcal{P}_l[\cdot] + \mathcal{P}_h[\cdot] = \mathcal{I}[\cdot], \quad (2.12)$$

$$\mathcal{P}_l[\mathcal{P}_l[\cdot]] = \mathcal{P}_l[\cdot], \quad \mathcal{P}_h[\mathcal{P}_h[\cdot]] = \mathcal{P}_h[\cdot], \quad (2.13)$$

$$\mathcal{P}_l[\mathcal{P}_h[\cdot]] = \mathcal{P}_h[\mathcal{P}_l[\cdot]] = 0 \quad (2.14)$$

where $\mathcal{I}[\cdot]$ is the identity operator.

Applying these operators to the incompressible Navier-Stokes equations yields the governing momentum equations for the two subspaces:

$$\frac{\partial \mathbf{U}_l}{\partial t} + \mathcal{P}_l[(\mathbf{U}_l \cdot \nabla) \mathbf{U}_l] = -\frac{1}{\rho} \nabla P_l + \nu \nabla^2 \mathbf{U}_l - \mathcal{P}_l[(\mathbf{u}_h \cdot \nabla) \mathbf{u}_h] \quad (2.15)$$

and

$$\frac{\partial \mathbf{u}_h}{\partial t} + \mathcal{P}_h[(\mathbf{u}_h \cdot \nabla) \mathbf{U}_l] + \mathcal{P}_h[(\mathbf{U}_l \cdot \nabla) \mathbf{u}_h] = -\frac{1}{\rho} \nabla p_h + \nu \nabla^2 \mathbf{u}_h, \quad (2.16)$$

where P_l and p_h are the respective pressure components enforcing $\nabla \cdot \mathbf{U}_l = 0$ and $\nabla \cdot \mathbf{u}_h = 0$. Note that within the QL framework any interaction between $\mathbf{U}_l$ and $\mathbf{u}_h$ generates modes strictly within the high-wavenumber subspace, consequently, applying the low-pass filter to these cross-terms yields exactly zero: $\mathcal{P}_l[(\mathbf{U}_l \cdot \nabla) \mathbf{u}_h] = \mathcal{P}_l[(\mathbf{u}_h \cdot \nabla) \mathbf{U}_l] = 0$. The core of the QL approximation lies in neglecting the fluctuation-fluctuation nonlinear scattering term, $\mathcal{P}_h[(\mathbf{u}_h \cdot \nabla) \mathbf{u}_h]$, in equation 2.16. Consequently, the high-wavenumber fluctuations are governed by linearised dynamics about the evolving low-wavenumber state, while the low-wavenumber state is driven by the high-wavenumber Reynolds stress tensor, $\mathcal{P}_l[(\mathbf{u}_h \cdot \nabla) \mathbf{u}_h]$. In standard QL studies [27; 14; 15; 12], the low-wavenumber subspace is strictly confined to the streamwise mean by setting $M_x = 0$ and $M_z = N_z$. This formulation will be referred to as QL_x hereafter.

Significance for the Self-Sustaining Process (SSP)

Extensive work by the Hwang group [28; 11] has demonstrated that the classical QL_x framework is sufficient to sustain turbulence and accurately reproduce the core turbulent features observed in DNS at low to moderate Reynolds numbers. This confirms that the QL framework perfectly isolates the active interactions of the classical SSP. Mathematically, this is evident in the partitioned equations. The low-wavenumber subspace $\mathbf{U}_l$ contains both the $(0, 0)$ mean shear flow and the $(0, k_z)$ streamwise streaks and vortices. Hence, equation (2.15) contains both the linear lift-up term, $\hat{v}(0, y, n\beta) \frac{\partial U(y)}{\partial y}$, to generate the streaks and the Reynolds stress term, $\mathcal{P}_l[(\mathbf{u}_h \cdot \nabla) \mathbf{u}_h]$, required for vortex regeneration from the interaction of the $\hat{\mathbf{u}}(\pm\alpha, 0)$ and the $\hat{\mathbf{u}}(\mp\alpha, \beta)$ modes. Concurrently, the interaction between the low- and high-wavenumber fields in equation (2.16) perfectly encapsulates the streak instability necessary to feed the fluctuating modes.

The Spanwise Quasilinear (QL_z) Framework

In this study, the QL framework is reconfigured to test the existence of an alternative pathway for turbulence sustenance by confining the low-wavenumber subspace entirely to the spanwise mean, setting $M_x = N_x$ and $M_z = 0$, a formulation referred to as QL_z hereafter. Under this geometry, the classical streamwise streaks and vortices are relegated to the high-wavenumber fluctuating subspace. Hence, the streak instability no longer drives the growth of the instability modes. Furthermore, because the QL approximation mathematically eliminates fluctuation-fluctuation scattering, $\mathcal{P}_h[(\mathbf{u}_h \cdot \nabla) \mathbf{u}_h] = 0$, the triadic interactions required to nonlinearly regenerate the $(0, \beta)$ streamwise vortices are completely erased, actively suppressing the traditional $k_x = 0$ SSP pathway. Therefore, if turbulence successfully sustains under the QL_z approximation, it rigorously demonstrates the existence of a novel, fundamentally distinct turbulence-maintenance process.

Chapter 3

Methodology

3.1 Computational Setup and Numerical Methodology

The numerical solver used in this study is `diablo` [29], an open-source pseudo-spectral code optimised for incompressible wall-bounded turbulent flows. The Navier-Stokes equations are discretised using Fourier series in the homogeneous streamwise and spanwise directions, while in the wall-normal direction the second-order central finite differences are applied on a staggered grid using an arctangent distribution to cluster grid points near the boundaries, ensuring the resolution of high near-wall velocity gradients. To compute the nonlinear convective terms, variables are transformed to physical space, multiplied, and transformed back to spectral space. Aliasing errors are strictly eliminated using the 2/3 rule, which truncates the upper one-third of wavenumbers prior to the nonlinear operations. The time integration is conducted semi-implicitly based on the fractional-step method [30]. Viscous terms are advanced implicitly with a second-order Crank-Nicolson scheme, while nonlinear advective terms are advanced explicitly using a low-storage third-order Runge-Kutta method.

The numerical simulations are executed exploiting the GQL framework as described in Section 2.3. For the QL cases, the cutoffs defining the $\mathcal{P}_l$ -subspace group are strictly chosen to be $M_x = 0$ for the QL_x configuration and $M_z = 0$ for the QL_z configuration, while for the DNS case they are chosen to be $M_x = N_x$ and $M_z = N_z$.

The computational domain simulates turbulent plane Couette flow by specifying the domain size and flow parameters as outlined in Section 2.1.1 and enforcing a zero pressure gradient throughout the channel. To initialise the simulations, the flow field is prescribed as the analytical laminar base state, $U = (y/h)U_w$, superimposed with random, finite-amplitude velocity perturbations to trigger transition to turbulence. For all simulations, the flow is advanced until statistically steady flow is reached, and the initial transient phase is discarded. The time step is chosen to allow for a CFL number lower than 0.4, and statistical parameters are then obtained by averaging over a time period of $10000 tU_w/h$, large enough to ensure convergence of all second-order statistics.

Chapter 4

Configuration and Benchmarking

4.1 Validation against Literature Data

To validate the numerical framework and ensure the physical fidelity of the simulated flow fields, the present DNS was benchmarked against two established databases of plane Couette flow. The first validation replicates the DNS of Kawahara & Kida [31] at $Re_w = 400$, while the second benchmark uses data from the Kawamura Laboratory database [32] at $Re_w = 12800$. For both cases, the spatial domains and physical parameters were replicated exactly. The grid resolution was strictly refined to compensate for the $2/3$ dealiasing rule and the numerical dissipation introduced by the second-order central finite-difference scheme applied in the wall-normal direction.

To confirm that the solver accurately captures the near-wall velocity gradients and the structural mechanisms driving Reynolds stress production, the mean streamwise velocity profiles and the RMS velocity fluctuations were compared against the reference data. As demonstrated in Figures 4.1 and 4.2, the comparisons yield excellent agreement, verifying the spatial resolution and the temporal integration scheme. This was further quantified using the relative L_2 norm error, defined as $\epsilon_{L_2} = \frac{\|\phi_{sim} - \phi_{ref}\|_2}{\|\phi_{ref}\|_2} \times 100\%$. As detailed in Table 4.1, the relative errors remain bounded within standard spectral tolerances, confirming that the structural dynamics of the SSP are rigorously captured.

Database	$\epsilon_{L_2}(U)$	$\epsilon_{L_2}(u_{rms})$	$\epsilon_{L_2}(v_{rms})$	$\epsilon_{L_2}(w_{rms})$
Kawahara & Kida ($Re_w = 400$)	1.63%	3.38%	5.76%	6.83%
Kawamura Lab ($Re_w = 12800$)	0.97%	1.56%	1.87%	2.28%

Table 4.1: Relative L_2 norm errors of mean and fluctuating velocity profiles compared to benchmark data.

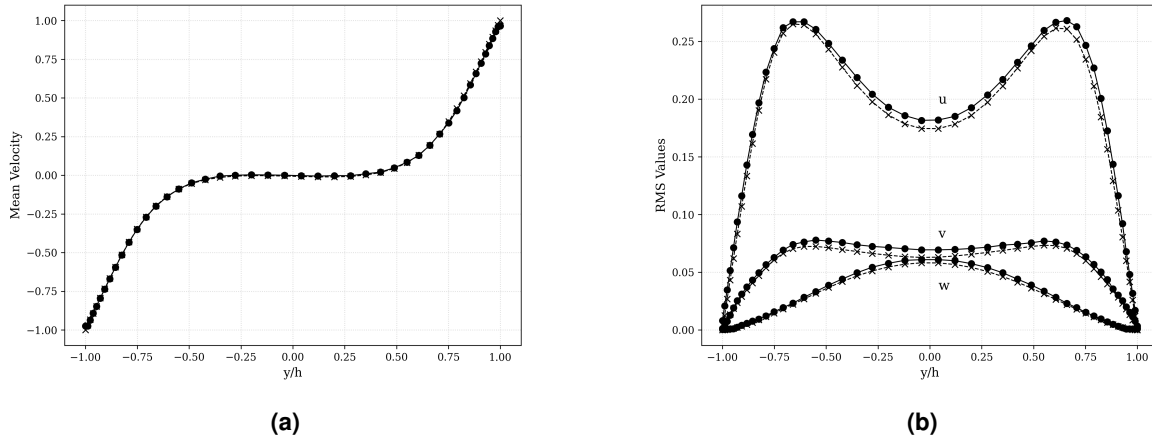


Figure 4.1: (a) Mean streamwise velocity profile, U , and (b) root-mean-square turbulent velocity fluctuations, u_{rms} , v_{rms} , and w_{rms} , comparing the present simulation (—, $\times$) with the reference simulation (---, $\circ$) at $Re_w = 400$

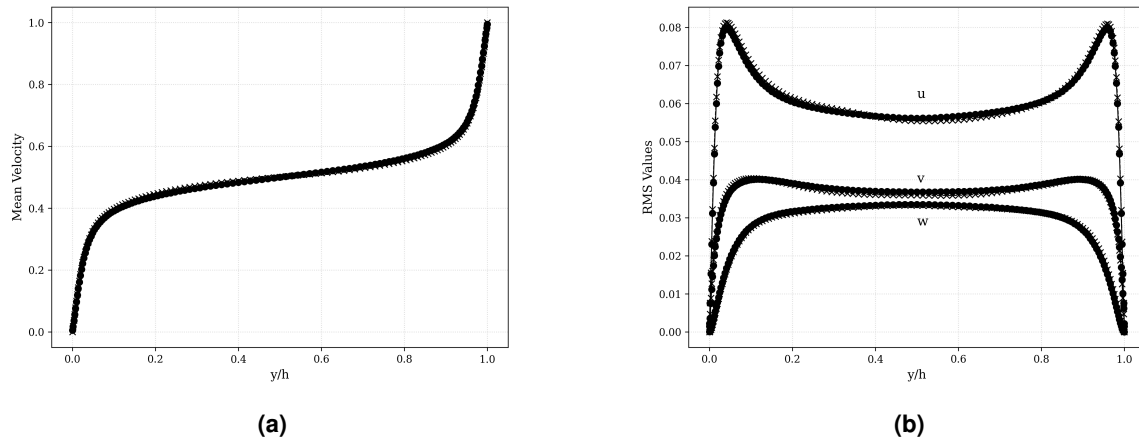


Figure 4.2: (a) Mean streamwise velocity profile, U , and (b) root-mean-square turbulent velocity fluctuations, u_{rms} , v_{rms} , and w_{rms} , comparing the present simulation (—, $\times$) with the reference simulation (---, $\circ$) at $Re_w = 12800$

4.2 Experiment Configuration

Three distinct numerical experiments were conducted: A fully resolved baseline DNS, a classical QL_x for comparison, and the object of the study, QL_z . The domain size was selected to isolate the minimal flow unit required to sustain large-scale motions at the specified Reynolds numbers to be $(L_x \times h \times L_z) = (10.9 \times 2 \times 5.5)$.

To ensure comparable turbulent statistics across all cases, the bulk Reynolds number was set to $Re_w = 1000$ for the DNS and QL_x simulations, and $Re_w = 1428$ for the QL_z simulation, yielding a consistent friction Reynolds number of $Re_\tau \approx 66, 65, 63$ respectively. The DNS spatial resolution was established with $84 \times 81 \times 84$ grid points, translating to near-wall spacings of $\Delta x^+ \approx 7 - 8$,

$\Delta z^+ \approx 4-5$, and $\Delta y^+ \approx 0.3-0.4$, and properly capturing the near-wall motions within the viscous sublayer [33]. For the QL_x and QL_z approximations, the linearised homogeneous direction was strictly truncated to 4 grid points. This ensures that only the mean and the fundamental spatial mode are retained after dealiasing, and physically forces turbulence maintenance to rely exclusively on the targeted low-wavenumber subspace interactions with the principal instability modes.

Chapter 5

Statistical Characteristics of QL_z

This chapter evaluates the statistics of turbulence maintained under QL_z . Figure 5.1 presents the mean-velocity profile, the streamwise and wall-normal turbulent velocity fluctuations, and the Reynolds shear stress for the DNS, QL_x , and QL_z configurations detailed in Section 4.2. The data demonstrates that a turbulent state is successfully sustained even in QL_z , indicating that a distinct, spanwise-driven self-sustaining process is operating there.

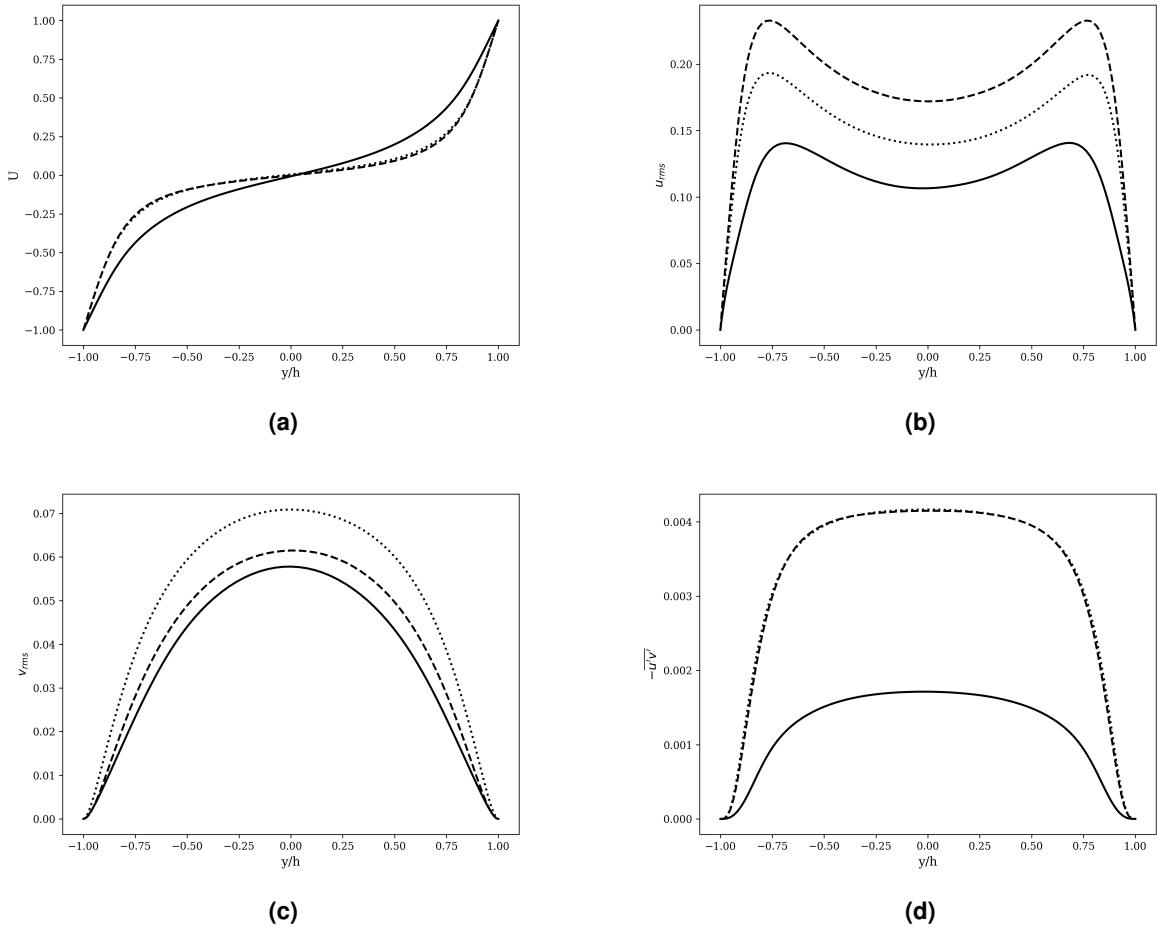


Figure 5.1: Wall-normal profiles of the turbulent statistics: (a) mean streamwise velocity, U ; (b) streamwise RMS velocity, u_{rms} ; (c) wall-normal RMS velocity, v_{rms} ; and (d) Reynolds shear stress, $-\langle u'v' \rangle$. The profiles compare the DNS baseline ($\cdots$), the standard QL_x results ($--$), and the QL_z results ($—$).

Analysis of the turbulent fluctuations reveals different energy distributions dictated by the respective spatial restrictions of the QL frameworks. As shown in Figure 5.1, the QL_x approximation overpredicts u_{rms} , agreeing with the behaviour reported in the literature [14; 15; 18]. It also underpredicts v_{rms} , but the combination of both results in a Reynolds stress that closely matches that of DNS. Consequently, QL_x accurately reproduces the DNS mean-velocity profile, as expected from the mean momentum balance equation 2.2.

Conversely, the QL_z framework yields a significantly weaker turbulent state. The Reynolds shear stress is largely underpredicted compared to DNS or QL_x , resulting in a less pronounced mean-velocity gradient at the wall. This reduction in turbulent momentum transfer can also be seen by the fact that a substantially higher bulk Reynolds number ($Re_w = 1428$) is required to achieve a friction Reynolds number ($Re_\tau \approx 63$) comparable to the DNS and QL_x cases.

Beyond the magnitude of the fluctuations, the QL_z approximation structurally modifies the near-wall turbulence dynamics, evidenced by the wall-normal shift of the u_{rms} peaks. In both the fully resolved DNS and the classical QL_x framework, the maximum streamwise velocity fluctuation occurs at $y^+ \approx 11$, corresponding to the near-wall streak distance from the wall [33]. The QL_x model captures this peak because its low-wavenumber subspace retains the full nonlinear interactions for all streamwise-constant modes $(0, k_z)$. Under the QL_z decomposition, however, the u_{rms} peak shifts outward to $y^+ \approx 18 - 20$. This shift is natural because only the fundamental $(0, \beta)$ streak exists in this framework. However, isolating the fundamental $(0, \beta)$ mode in DNS also reveals an outward peak shift to $y^+ \approx 25 - 27$, consistent with Townsend's attached eddy scaling [11]. The fact that the QL_z peak remains distinct confirms that the streaks sustained under QL_z are structurally distinct from classical near-wall streaks.

Chapter 6

Dynamics of the Spanwise-Driven SSP

This section investigates the dynamics of the spanwise-driven SSP maintained under the QL_z approximation. To track the behaviour of the coherent structures governing this cycle, we first define the domain-averaged turbulent kinetic energy for each fluctuating velocity component over a small domain around the centreline spanned in the wall-normal direction by a length L_y :

$$E_u(t) = \frac{1}{V} \int_V u'^2 dV, \quad E_v(t) = \frac{1}{V} \int_V v'^2 dV, \quad E_w(t) = \frac{1}{V} \int_V w'^2 dV, \quad (6.1)$$

where $V = L_x \times 2L_y \times L_z$ is the total volume of the domain. As established in the classical SSP framework, quasi-streamwise vortices are rotational structures embedded in the cross-stream ($y - z$) plane. Consequently, they are inherently related to the wall-normal (E_v) and spanwise (E_w) kinetic energies. For this reason, they will hereafter be collectively referred to as the vortex energy.

To isolate the specific triadic interactions driving the flow, we also define the kinetic energy of individual spatial Fourier modes. Following the decomposition introduced in Section 2.1.2, the energy of a specific mode is defined as:

$$1/L_y \int_0^{L_y} |\hat{u}(y; k_x, k_z)^2 + \hat{v}(y; k_x, k_z)^2 + \hat{w}(y; k_x, k_z)^2| dy \quad (6.2)$$

Using this formulation, three physically significant energy modes are tracked: the energy representing the classical spanwise streaks, E_s , defined by the fundamental streamwise-constant mode $(0, \beta)$, the energy of the principal instability, E_i , defined by the mode (α, β) , and the energy of the fundamental wavy spanwise-mean mode E_x , defined by $(\alpha, 0)$.

Figure 6.6 presents the time evolution of E_s , E_x , and E_v for both the fully resolved DNS and the QL_z configuration.

In the DNS, E_s and E_v exhibit a clear oscillation cycle characteristic of the classical SSP, which has a period of approximately $100 th/U_w$, in agreement with Hamilton et al. [1].

In contrast, the QL_z system displays a stronger correlation between E_s and E_x , and the oscillation period of the cycle seems to be around $50 th/U_w$. Furthermore, the reduced energy gap between E_s and E_x in QL_z compared to DNS indicates that the streaks might have a less important role in the energy-extraction pathway sustaining the turbulent state.

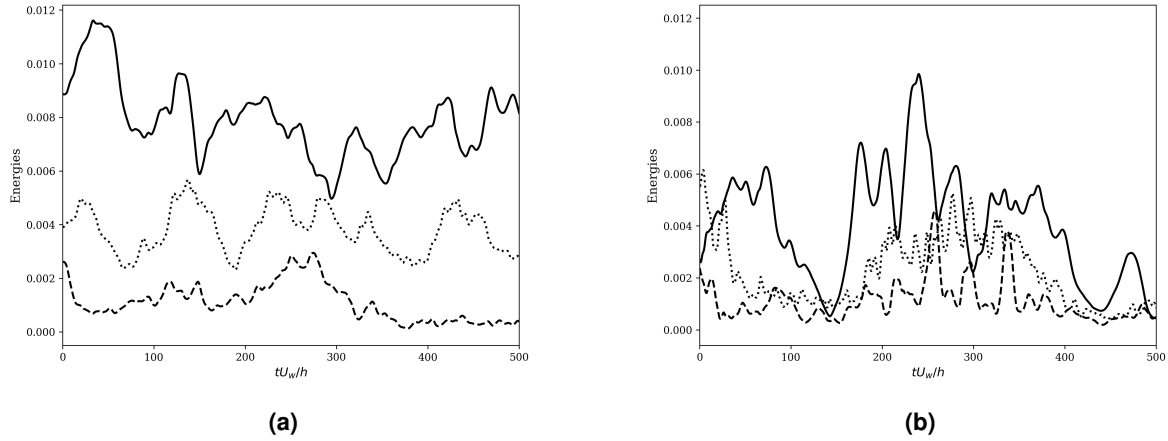


Figure 6.1: Time evolution of the streak energy (—), wavy spanwise-mean mode energy (— —) and the vortex energy ($\cdot \cdot \cdot$) for DNS, (a), and QL_z , (b)

6.1 Base Flow Stability Analysis

The fundamental mechanism of the classical SSP relies on the linear transient growth of streaks and their subsequent breakdown to excite turbulent fluctuations. To determine whether a dynamically analogous instability operates within the novel spanwise-driven QL_z framework, we must investigate whether the spanwise-averaged base flow is unstable and capable of driving the high-wavenumber fluctuations.

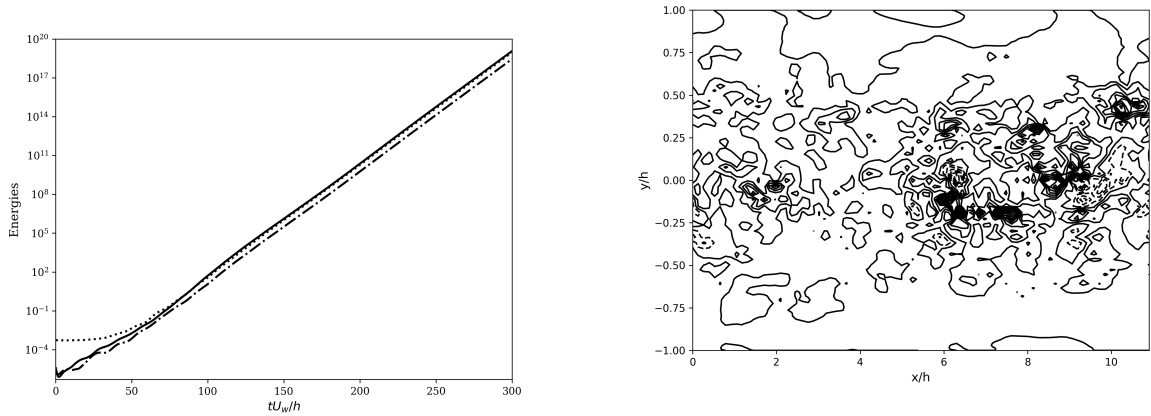
To evaluate this, a linear stability analysis was conducted. The base flow was extracted from the QL_z simulation at $tU_w/h = 268$, corresponding to a local maximum in the kinetic energy of the wavy spanwise-mean structure. To isolate the linear dynamics, the $\mathcal{P}_l$ -subspace, which contains the $(0, 0)$ mean shear and the $(\alpha, 0)$ fundamental wavy mode, was artificially frozen in time. The remaining high-wavenumber velocity modes were replaced with random, small-amplitude perturbations uniformly distributed between 0 and $10^{-5}U_w$.

Effectively, this serves as a method to extract the leading eigenmode of the linearised system. Note that because the base flow is periodic in x , the instability modes must take the form of $e^{i\beta z} \sum_n c_n e^{in\alpha x}$. Hence, the linearised Navier-Stokes operator couples all streamwise harmonics for a given spanwise wavenumber β . Therefore, if the flow is unstable, all constituent x -harmonics of the $\mathcal{P}_h$ -subspace grow at the identical exponential rate because they belong to the same global eigenmode.

Since the $(\alpha, 0)$ mode heavily dominates the energy of the wavy spanwise-mean flow, this computational experiment directly tests whether the strong shear layers of this z -independent structure are unstable. Figure 6.2a confirms this hypothesis, demonstrating the exponential growth of the $\mathcal{P}_h$ fluctuations and verifying that the wavy spanwise-mean flow is a dynamically essential structure in the spanwise-driven SSP, undergoing a structural breakdown analogous to classical streak instability, which subsequently energises the fluctuating modes.

Visualising the most excited eigenmodes in the $y - z$ plane, and initialising the linear instability

from different temporal peaks of the $(\alpha, 0)$ energy reveals that different base states excite different eigenmodes. This indicates that the spanwise-mean flow evolves on a timescale comparable to the instability growth rate itself. Consequently, isolating the most excited eigenmode holds no physical significance, closely mirroring the transient, quasi-steady nature of classical streak instability where the base state is continuously modified by the nonlinear regeneration cycle [1]. An example of the dominant streamwise velocity eigenmode extracted from the linear instability of the $(\alpha, 0)$ base state visualised in the $y - z$ plane, is shown in Figure 6.2b.



(a) Linear stability of $k_z = 0$ modes of flow from full simulation at $tU_w/h = 268$: streak energy (—), principle instability energy (— · —) and vortex energy (· · · ·)

(b) Iso-contours of u in the (x, y) -plane at $z/h = 2$ with contour interval $0.1U_w$, where solid lines are positive values and dashed lines are negative values, showing the shape of the dominant eigenmode

Figure 6.2

6.2 Triadic Interactions

Having established that the $(\alpha, 0)$ velocity mode drives the high-wavenumber fluctuations, we must now determine the physical mechanisms responsible for feeding energy back into the energy of the wavy spanwise-mean itself. According to the spectral kinetic energy budget outlined in Section 2.1.2, energy transfer into this specific wavenumber can only occur via linear production by extraction from the mean shear, $\hat{P}(\alpha, y, 0)$, or turbulent transport, $\hat{T}_{turb}(\alpha, y, 0)$. Within the QL_z formulation, this transfer is driven strictly by the Reynolds stress tensor, $\mathcal{P}_l[(\mathbf{u}_h \cdot \nabla)\mathbf{u}_h]$, in the low-wavenumber momentum equation. The dominant triadic interactions generating this stress involve the fundamental streak and instability modes, $(0, \pm\beta)$ and $(\alpha, \mp\beta)$, alongside cross-interactions with higher harmonics such as $(-\alpha, \pm\beta)$ and $(2\alpha, \mp\beta)$.

To evaluate the relative contribution of these mechanisms, the temporal evolution of both $\hat{P}(\alpha, y, 0)$ and the main contributions to $\hat{T}_{turb}(\alpha, y, 0)$ were computed. As shown in Figure 6.3, the energy injected into the wavy spanwise-mean mode is dominated by turbulent transport, with direct mean extraction playing a negligible role.

This demonstrates that the $(\alpha, 0)$ structure does not grow via a linear interaction with the $(0, 0)$ mean flow, but is dynamically sustained by the continuous nonlinear scattering of the $\mathcal{P}_h$ fluc-

tuations, serving as the structural analogue to the vortex-regeneration phase of the classical SSP.

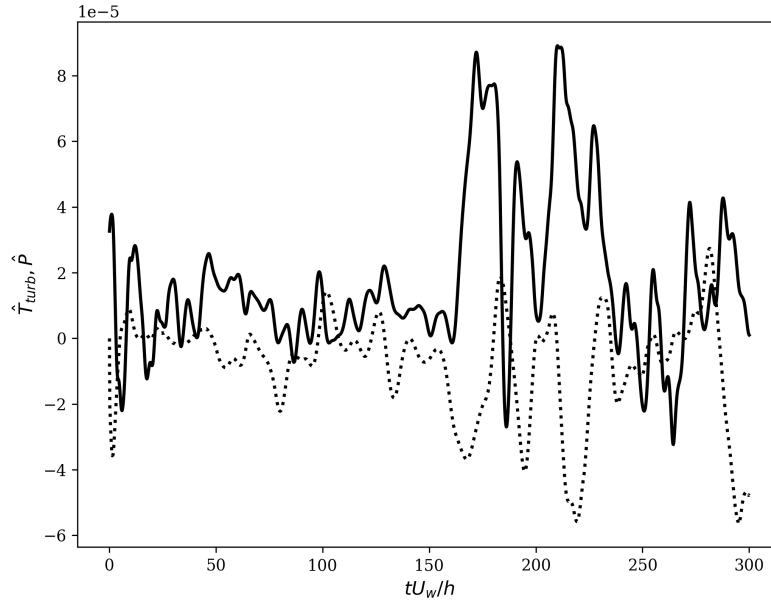


Figure 6.3: Time evolution of the $\hat{T}_{turb}(\alpha, y, 0)$ component (—), and $\hat{P}(\alpha, y, 0)$ component (· · · · ·) energising $(\alpha, 0)$ energy mode

6.3 Lift-up and Vortex Generation in QL_z

Turbulent fluctuations cannot be sustained solely by linear energy redistribution; they must continuously extract energy from the mean shear to overcome viscous dissipation. Visual inspection of the flow cross-section during peak E_s , presented in Figure 6.4, reveals that classical spanwise streaks persist within the spanwise-driven SSP. These structures actively extract energy from the mean flow via the linear lift-up effect. Mathematically, this is permitted because the necessary linear production term, $\hat{v}(0, y, n\beta) \frac{\partial U(y)}{\partial y}$, resides within the high-wavenumber $\mathcal{P}_h$ -subspace of the QL_z equations.

However, the mechanism governing the generation of quasi-streamwise vortices in this configuration diverges fundamentally from the classical SSP. In the classical cycle, streamwise vortices are nonlinearly regenerated by the Reynolds stress resulting from the triadic interactions of the principal streak instability modes. However, this term is explicitly neglected under the QL_z approximation. This suggests that they are products of the primary instability wavepacket driven by the wavy spanwise-mean flow.

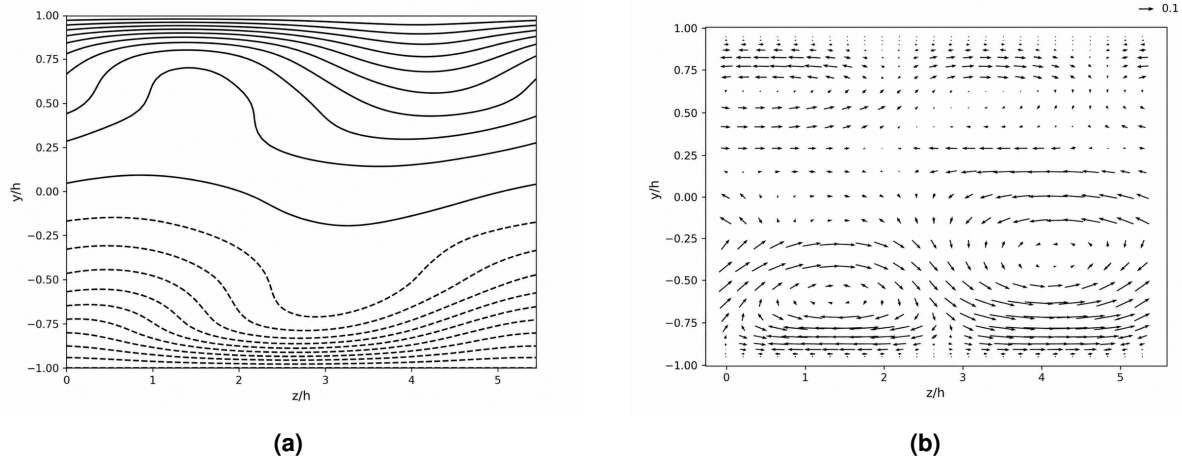


Figure 6.4: Streamwise (a) and cross-flow (b) velocities in the (y, z) -plane at $tU_w/h = 241$. (a) shows iso-contours of u with contour interval $0.1U_w$ where solid lines are positive values, dashed lines are negative values, and (b) shows the velocity vectors for (v, w) with a reference vector of magnitude 0.1

6.4 Isolation of Active Triads

To definitively identify the minimal set of modes required to close the spanwise-driven regenerative loop, a series of targeted numerical experiments was conducted in which specific Fourier modes were artificially zeroed at every time step, and the macroscopic response of the turbulent state was observed.

The first experiment tested the necessity of the fundamental wavy spanwise-mean structure by zeroing the $(\alpha, 0)$ mode. Upon suppression, the turbulent kinetic energy underwent a slow viscous decay. The high-wavenumber oblique fluctuations, $(\alpha, \pm\beta)$, and the quasi-streamwise vortices, (v', w') , rapidly dissipated, followed by the slow viscous decay of the $(0, \beta)$ streamwise streaks over approximately $800h/U_w$. This explicitly demonstrates that the $(\alpha, 0)$ spanwise rollers are mathematically necessary to drive the linear instability that sustains the fluctuating field, as postulated previously.

The second experiment targeted the $(\alpha, \pm\beta)$ oblique modes. Suppressing these modes similarly resulted in the total viscous decay of the system. This isolates the $(\alpha, \pm\beta)$ structures as the mandatory nonlinear feedback mechanism. Without their presence, the system lacks the required Reynolds stress divergence to sustain the $(\alpha, 0)$ spanwise rollers against viscous dissipation, thereby breaking the triadic interaction loop.

The final experiment evaluated the role of the canonical $(0, \beta)$ streamwise streaks. Strikingly, when these modes were truncated, the turbulent state completely survived. The continuous interaction between the $(\alpha, 0)$ and $(\alpha, \pm\beta)$ modes with higher harmonics was mathematically sufficient to independently maintain the flow. This result represents a fundamental departure from the classical SSP, where the streak instability is the main engine of turbulence maintenance. In the QL_z framework, the classical streaks are revealed to be dynamically secondary. While they extract significant energy from the mean flow via the linear lift-up effect and contribute to

the total turbulent transport feeding the $(\alpha, 0)$ mode, they are not strictly required to close the self-sustaining cycle. Instead, the $(\alpha, \pm\beta)$ oblique modes extract sufficient energy directly from the mean shear and generate the necessary Reynolds stress divergence to sustain the $(\alpha, 0)$ structure, thereby closing the spanwise-driven SSP and entirely bypassing the canonical $k_x = 0$ pathway.

To further investigate the energy source of the oblique modes, the spectral production term, $\hat{P}(\alpha, y, \beta)$ was computed as a function of time for a normally developing QL_z flow. As demonstrated in Figure 6.6b, the $(\alpha, \pm\beta)$ modes continuously extract kinetic energy from the mean flow, with $\hat{P}(\alpha, y, \beta)$ oscillating to a peak amplitude of 5×10^{-4} . This production mechanism is much larger than the same instability's contribution to turbulent transfer detailed in Figure 6.3 and thus dominates its energy balance. However, the mechanism by which this oblique mode extracts energy from the mean is unknown.

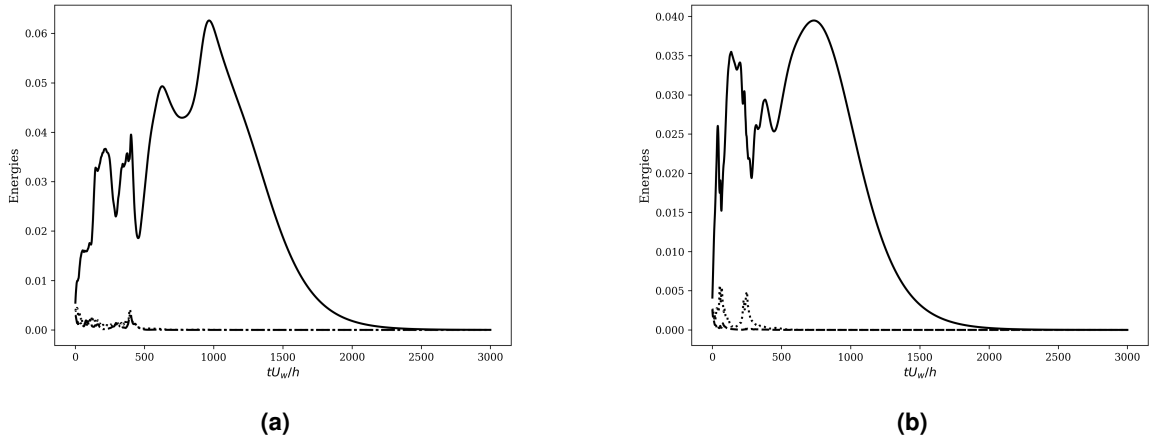
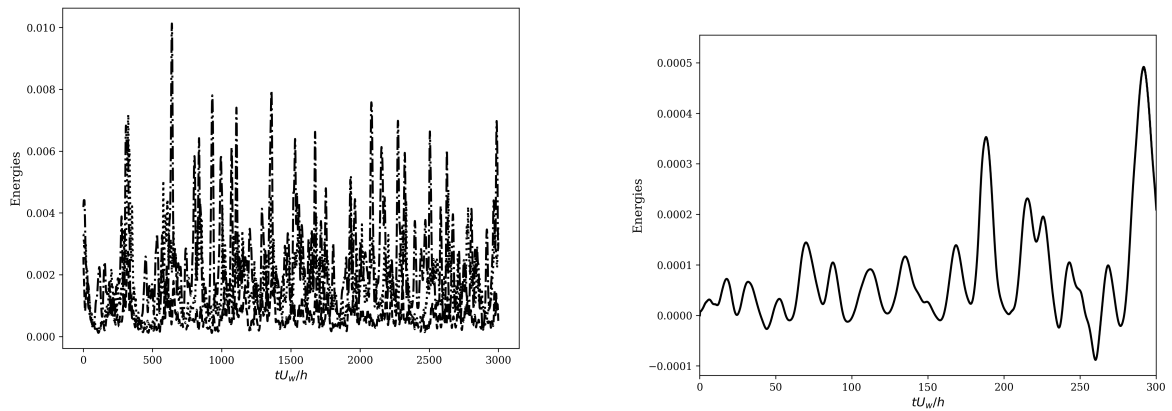


Figure 6.5: Time evolution of E_s (—), E_x (---), E_v (·····) and E_i for the QL_z framework under targeted modal truncation: zeroed E_x (a) and zeroed E_i (b)



(a) Time evolution of E_x (---), E_v (·····) and E_i for QL_z with zeroed E_s

(b) Time evolution of $\hat{P}(\alpha, y, \beta)$ for the QL_z framework

Figure 6.6

Chapter 7

Discussion

7.1 Spanwise-driven SSP

The previous analyses delineate the dynamics of the novel, spanwise-driven self-sustaining process operating under the QL_z approximation. In contrast to the classical SSP dominated by streamwise coherent structures, this alternative cycle maintains turbulence via a feedback loop that entirely bypasses the classical $k_x = 0$ pathway. First, the oblique modes, $(\alpha, \pm\beta)$, and the streaks, $(0, \beta)$, extract kinetic energy directly from the $(0, 0)$ mean shear. Following energy extraction, the streaks together with the oblique modes, and the oblique modes together with higher harmonics interact to generate the Reynolds stress required to transfer energy into the fundamental wavy spanwise-mean structure, $(\alpha, 0)$. Energised strictly by this turbulent transport, the $(\alpha, 0)$ spanwise rollers amplify until the structure supports a linear instability. The subsequent structural breakdown of this wavy spanwise-mean base flow directly excites the fluctuations and vortices required to support the production of oblique modes and streaks. This closes the loop, continuously sustaining the turbulent state. The dynamics of this spanwise-driven SSP are presented schematically in the following diagram 7.1.

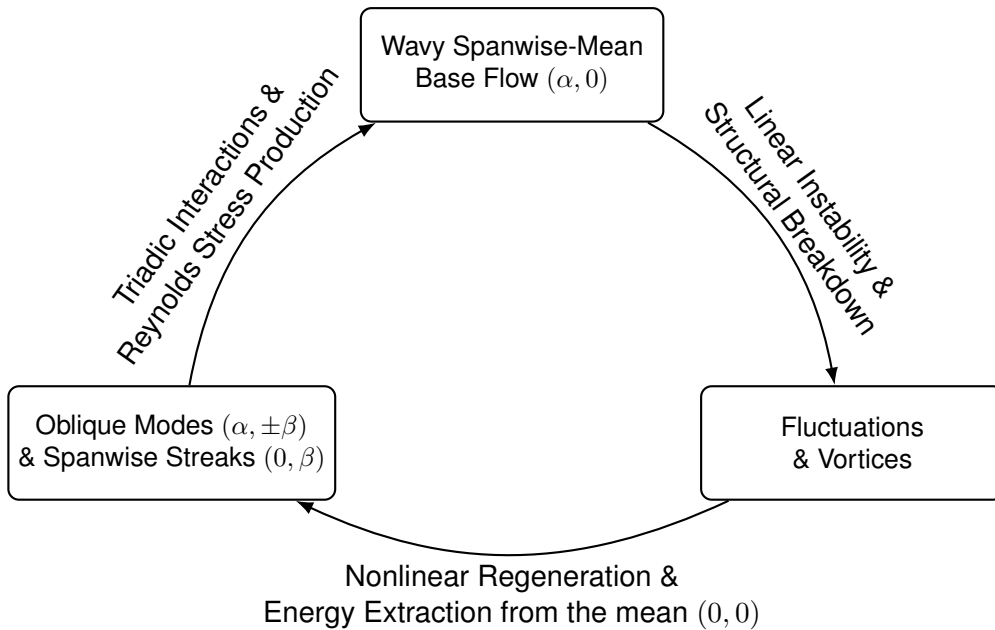


Figure 7.1: Schematic representation of the spanwise-driven SSP under the QL_z approximation

The spanwise-driven self-sustaining process represents a fundamental departure from the canonical turbulence maintenance cycle. In the classical SSP, turbulence relies on a continuous $k_x = 0$ pathway: quasi-streamwise vortices generate spanwise alternating streaks via the linear lift-up effect, which subsequently break down through a streak instability to nonlinearly regenerate the vortices. The QL_x framework mathematically captures this mechanism by retaining the $(0, k_z)$ low-wavenumber subspace, allowing the streak instability to excite fluctuating modes whose triadic interactions drive vortex regeneration. Instead, the primary engine of turbulence maintenance under the QL_z formulation shifts to the $(\alpha, \pm\beta)$ oblique modes. These modes extract kinetic energy directly from the $(0, 0)$ mean shear. The classical cycle's reliance on streak instability is consequently replaced by the structural breakdown of the fundamental wavy spanwise-mean mode, $(\alpha, 0)$. Furthermore, while the classical cycle relies on the streak to extract energy from the mean and feed it to the rest of the fluctuations, they are dynamically secondary within the QL_z framework, where it is mainly the oblique mode that extracts energy from the mean, distributes it to the wavy spanwise-mean, which then excites the rest of the fluctuations.

Chapter 8

Conclusion

8.1 Summary of Findings

This study has established the existence of a novel, spanwise-driven self-sustaining process in turbulent plane Couette flow. By applying a spanwise quasilinear (QL_z) approximation, the low-wavenumber subspace was strictly confined to the spanwise mean. This mathematical configuration successfully isolated a previously undocumented turbulence maintenance mechanism that operates independently of the canonical $k_x = 0$ pathway established by Hamilton, Kim, & Waleffe [1].

The classical structural view of wall-bounded turbulence relies on the dominance of streamwise coherent structures. In the traditional SSP, the linear transient growth of spanwise streaks, their breakdown, and the subsequent nonlinear scattering are required to regenerate quasi-streamwise vortices. Under the QL_z framework, these canonical streamwise streaks and vortices were relegated to the high-wavenumber subspace. Because the quasilinear approximation explicitly eliminates fluctuation-fluctuation nonlinear scattering ($\mathcal{P}_h[(u_h \cdot \nabla)u_h] = 0$), the classical triadic interactions responsible for vortex regeneration were actively mathematically suppressed. Despite the elimination of the classical nonlinear feedback loop, the experiments successfully maintained a statistically steady turbulent state at $Re_\tau \approx 63$. This proved that the classical streamwise pathway is not the exclusive mechanism for turbulence sustenance. Instead, spectral energy budgets and targeted mode-suppression experiments revealed a fundamentally distinct mechanism.

The spanwise-driven SSP is governed by the wavy spanwise-mean structure, specifically the $(\alpha, 0)$ mode. This structure replaces the classical streak as the primary driver of the cycle. Linear stability analysis confirmed that the base flow comprising the mean flow and the $(\alpha, 0)$ mode is highly unstable, breaking down to energise the high-wavenumber fluctuating field. Crucially, truncation experiments demonstrated that, while canonical $(0, \beta)$ streaks continue to form via the linear lift-up effect, remain highly energetic and feed energy to the wavy spanwise-mean, they are dynamically secondary, as the turbulence survives without them. Instead, the regenerative loop is successfully closed solely by the $(\alpha, \pm\beta)$ oblique modes. These oblique structures extract sufficient kinetic energy directly from the mean shear and sustain the $(\alpha, 0)$ spanwise rollers through Reynolds stress forcing.

The discovery of this alternative, spanwise-driven maintenance mechanism might reveal a missing physical component in the theoretical framework of shear turbulence which may operate

alongside the classical SSP.

8.2 Future Work

Several fundamental physical mechanisms require further investigation.

First, the exact structural nature of the quasi-streamwise vortices must be formally classified. Because the traditional fluctuation-fluctuation scattering term, $\mathcal{P}_h[(u_h \cdot \nabla)u_h]$, is explicitly truncated in the quasilinear approximation, the canonical vortex-regeneration mechanism is mathematically suppressed. The present results suggest that the observed vortices are merely geometric byproducts of the three-dimensional instability wavepacket driven by the $(\alpha, 0)$ mode, but future research should investigate this further.

Furthermore, the truncation experiments established that the $(\alpha, \pm\beta)$ oblique modes act as the mandatory engine of the spanwise-driven cycle, extracting sufficient energy from the mean shear. Future efforts should provide a mathematical description of what the nature of the mean shear energy extraction is.

Moreover, the present investigation is restricted to low Reynolds numbers ($Re_\tau \approx 63$). Future QL_z simulations must extend to higher friction Reynolds numbers.

Finally, and most importantly, it is left for future work to quantify its importance with respect to wall-bounded turbulence maintenance.

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